Chat with Energy Documents (RAG)

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement

The energy sector generates a vast amount of unstructured data in the form of technical reports, maintenance manuals, research papers, and operational documents. These documents contain critical insights required for decision-making, troubleshooting, and analysis. However, due to their large size and complex structure, extracting relevant information manually becomes a time-consuming and inefficient task.

Traditional search techniques, such as keyword-based search, are limited in their ability to understand the context of a query. They often return irrelevant or incomplete results because they rely only on exact word matches rather than semantic meaning. This creates a challenge for professionals who need precise and context-aware information quickly.

Additionally, manually going through hundreds of pages to find specific answers leads to reduced productivity and increased chances of human error. There is also a lack of intelligent systems that can interpret domain-specific knowledge in the energy sector effectively.

To address these challenges, there is a need for an advanced solution that can understand natural language queries, retrieve relevant information from large document repositories, and generate accurate responses. This project proposes the development of a Retrieval-Augmented Generation (RAG) system that combines semantic search with generative AI to provide efficient and context-aware answers based on uploaded energy documents.

2.Project Objectives

The primary objective of this project is to design and develop an intelligent document interaction system using Retrieval-Augmented Generation (RAG). The system aims to bridge the gap between traditional document search methods and modern AI-driven solutions by enabling users to interact with energy-related documents in a more efficient and meaningful way.

One of the key objectives is to allow users to upload energy-related documents such as PDFs and extract meaningful information from them. The system is designed to process unstructured text data and convert it into structured knowledge using advanced natural language processing techniques.

Another important objective is to implement semantic search using vector embeddings. Unlike traditional search systems, this approach focuses on understanding the meaning of the query and retrieving the most relevant information based on context.

The project also aims to integrate Large Language Models (LLMs) to generate humanlike responses. These models use the retrieved document context to provide accurate and relevant answers, ensuring that the output is grounded in the actual content of the documents.

Furthermore, the system is designed to improve efficiency by reducing the time required for manual document analysis. It also aims to provide scalability so that it can handle large volumes of data and be extended to other domains in the future.

Overall, the project focuses on enhancing productivity, improving accuracy, and providing an intelligent interface for document-based question answering.

3.Scope of the Project

The scope of this project includes the design and implementation of a Retrieval Augmented Generation (RAG) system specifically tailored for energy-related documents. The system is capable of handling unstructured data in the form of PDF files and transforming it into a searchable and interactive knowledge base.

The project focuses on document ingestion, where users can upload multiple documents into the system. These documents are then processed through various stages, including text extraction, chunking, and embedding generation. The processed data is stored in a vector database, enabling efficient semantic search.

Another important aspect of the project scope is the query processing mechanism. Users can input natural language queries, and the system retrieves relevant information from the stored document embeddings. The retrieved data is then passed to a Large Language Model, which generates a context-aware response.

The system is designed to be scalable and can be extended to support additional file formats such as Word documents and spreadsheets. It can also be adapted for use in other industries such as healthcare, finance, and legal services, where document analysis plays a crucial role.

However, the current scope is limited to text-based documents and does not include image or video processing. The system also relies on pre-trained language models and does not involve training custom models from scratch.

In summary, the project provides a strong foundation for intelligent document processing and can be expanded further with additional features and capabilities.

4. Proposed Solution A. Document Processing Phase

1. The system begins with **document ingestion**, where users upload energy-related documents in PDF format.
2. The uploaded documents undergo **text extraction**, where the content is converted into machine-readable text using specialized libraries.
3. The extracted text is then divided into **smaller chunks**, which helps in improving the efficiency and accuracy of information retrieval.
4. Each text chunk is converted into a **vector embedding**, which represents the semantic meaning of the text using numerical values.
5. These embeddings are stored in a **vector database**, which allows efficient similarity-based search operations.

B. Query Processing Phase

1. When a user enters a query, it is processed as a **natural language input**, making the system user-friendly and intuitive.
2. The query is converted into an embedding using the same embedding model used for document chunks.
3. The system performs a **similarity search** in the vector database to find the most relevant chunks related to the query.
4. The retrieved chunks are then passed as **context** to a Large Language Model (LLM).
5. The LLM generates a **context-aware response**, ensuring that the answer is accurate and relevant to the query.

C. Advantages of the Solution

1. Combines **retrieval and generation**, improving accuracy
2. Reduces irrelevant search results
3. Provides **context-based answers** instead of generic responses
4. Enhances user experience through natural language interaction

5. Key Features

Document Upload Capability :-

Allows users to upload multiple PDF documents Simplifies document management and processing

Efficient Text Chunking :-

Breaks down large documents into smaller segments o Improves search accuracy and retrieval speed

Semantic Search Mechanism :-

Uses vector embeddings to understand context Provides more relevant results compared to keyword search

Integration with LLMs :-

Generates human-like responses Ensures answers are meaningful and contextually accurate

Real-Time Query Processing :-

Provides fast and efficient responses Enhances user experience

Accuracy and Reliability :-

Ensures responses are based only on uploaded document

Reduces misinformation

Scalable Architecture :-

Can handle large datasets Easily extendable to other domains

6. Overall Architecture / Workflow

The system begins with the document upload stage, where users provide input data in the form of PDF files containing energy-related information such as reports, manuals, or research documents. This step acts as the entry point of the system and ensures that the data is available for further processing.

Once the document is uploaded, it undergoes a text extraction process, where specialized libraries are used to extract textual content from the PDF files. This step converts unstructured document data into raw text format, making it suitable for further analysis and processing by the system.

After text extraction, the system performs data preprocessing, which may include removing unnecessary characters, handling formatting issues, and normalizing the text. This step ensures that the extracted data is clean, consistent, and ready for efficient processing.

The processed text is then divided into smaller segments using text chunking techniques. Chunking is important because large documents cannot be processed efficiently as a whole. By splitting the text into manageable chunks, the system improves retrieval accuracy and ensures better performance during similarity search.

Each text chunk is then transformed into a vector embedding, which is a numerical representation of the text. These embeddings capture the semantic meaning and contextual information of the text, enabling the system to understand relationships between different pieces of information.

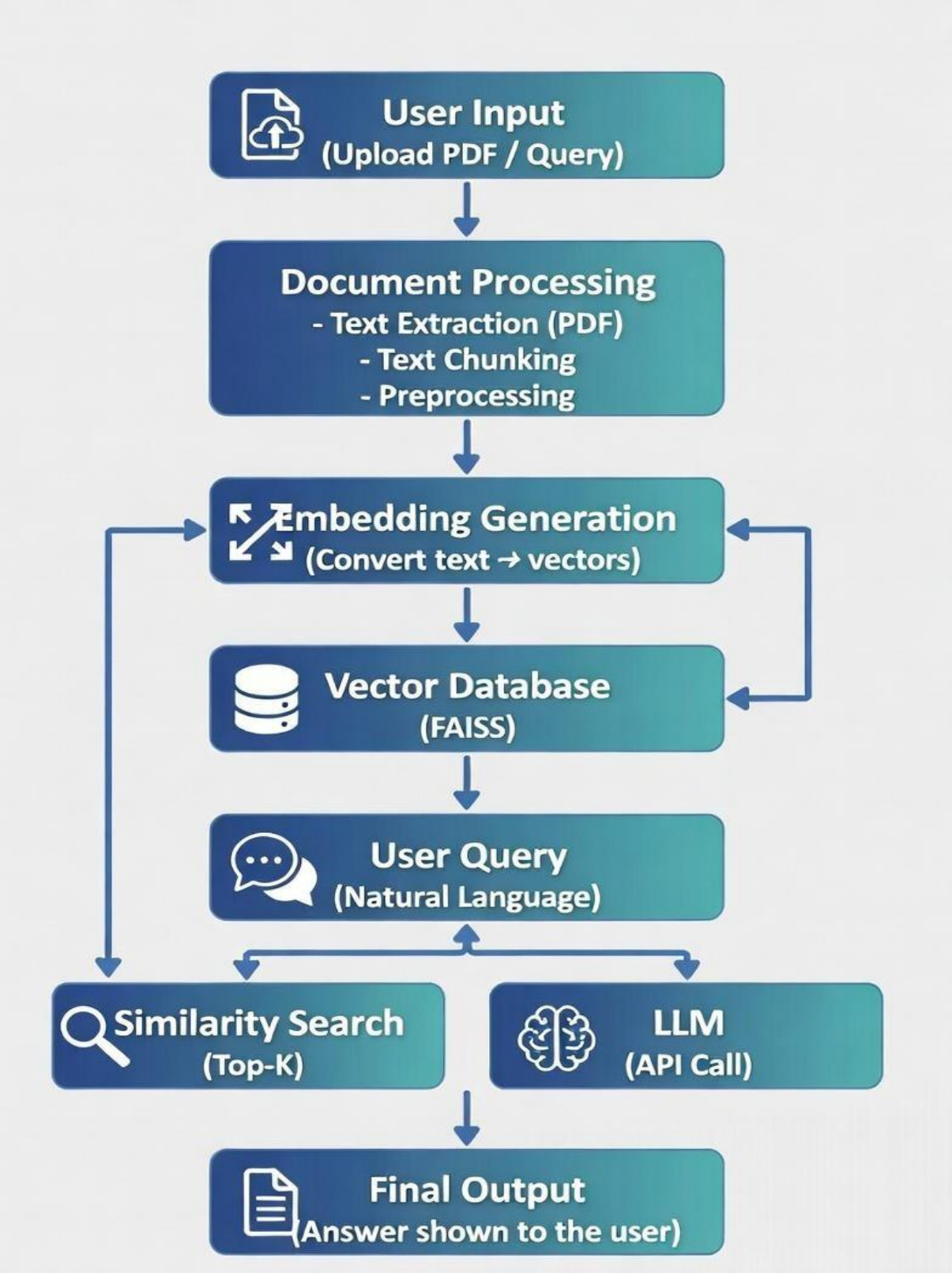
The generated embeddings are stored in a vector database, such as FAISS or Pinecone. This database is optimized for handling high-dimensional data and allows fast and efficient similarity-based searches. It acts as the knowledge base of the system.

When a user submits a query, the system first processes it as a natural language input, making the interaction simple and user-friendly. The query is then converted into an embedding using the same model used for document chunks.

The system performs a similarity search operation by comparing the query embedding with the stored embeddings in the vector database. This process identifies the most relevant document chunks based on semantic similarity rather than keyword matching.

The top relevant chunks retrieved from the database are then passed as contextual input to the Large Language Model (LLM).

Fig.1->RAG System Architecture Diagram (Main Diagram)



7. Tools & Technologies Used

- **Python**

  Used as the primary programming language Provides flexibility and extensive library support

- **Lang Chain**

  Used to build the RAG pipeline Manages document processing and retrieval

- **OpenAI / LLM API**

  Used for generating responses Provides natural language understanding

- **Py PDF**

  Used for extracting text from PDF files

- **NumPy**

  Used for numerical computations

- **Vector Databases (FAISS / Pinecone / ChromaDB)**

  Store embeddings efficiently Enable fast similarity search

- **Streamlit**

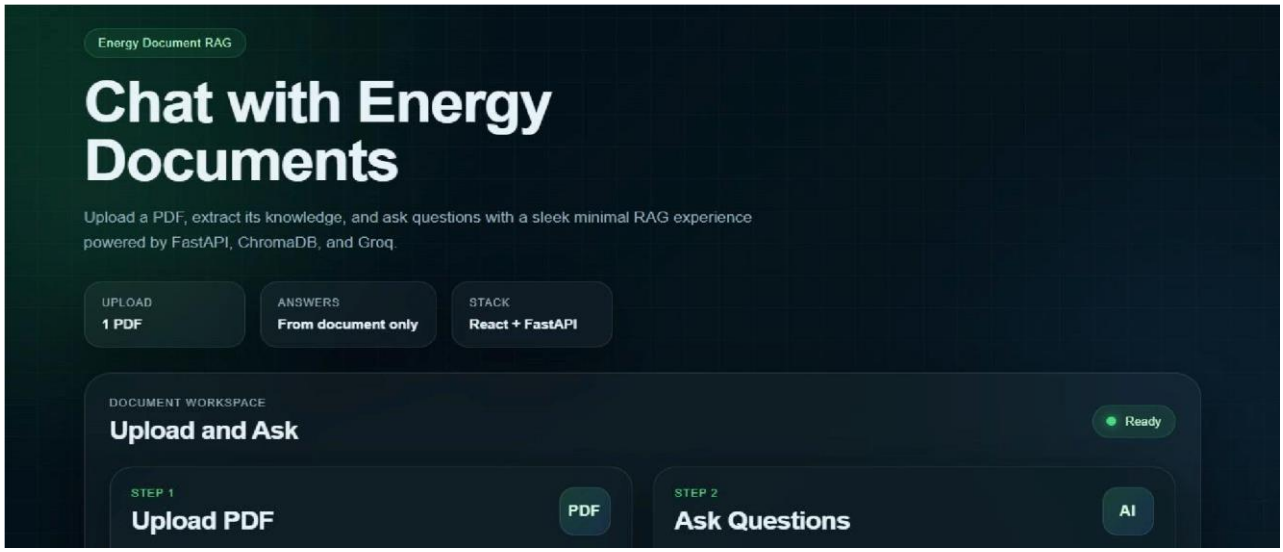
  Used for building a simple and interactive user interface

- **VS Code**

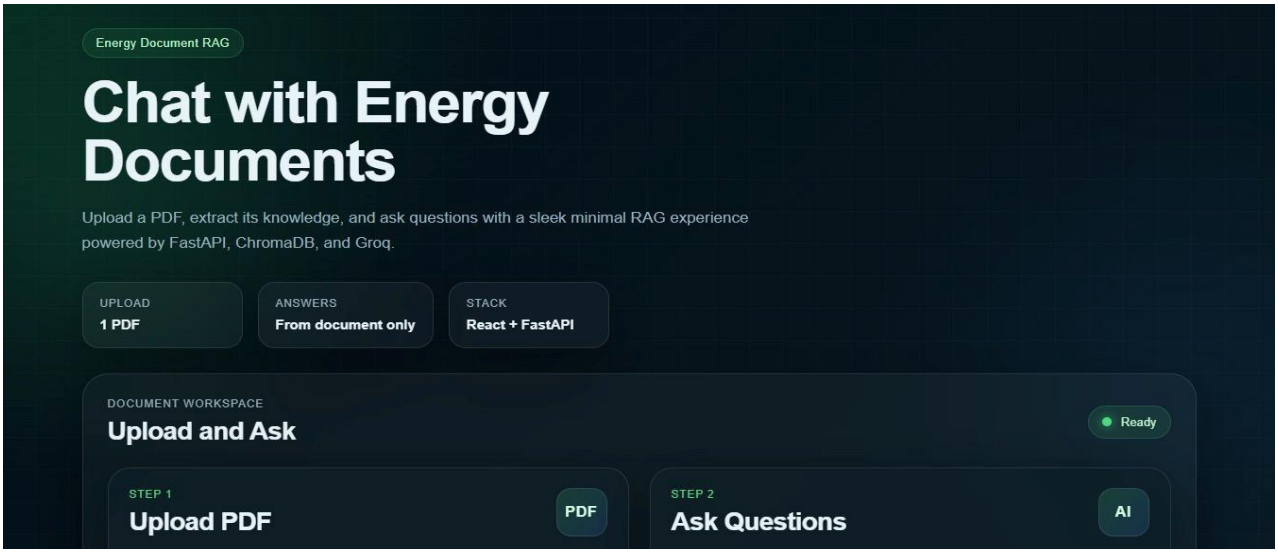
  Used as the development environment

8. Results & Output

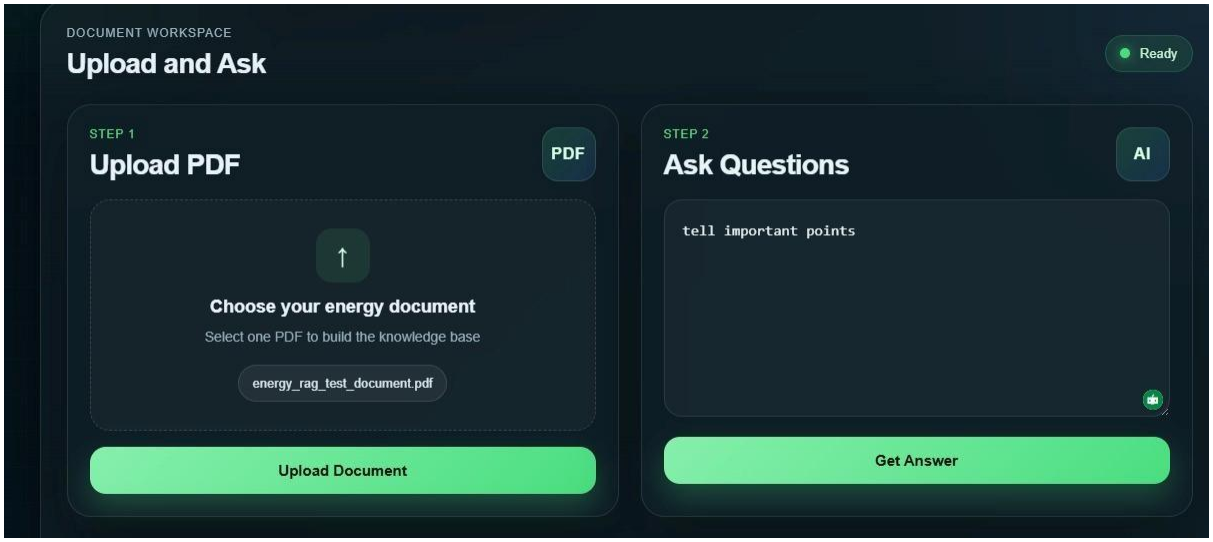
A . Landing Page



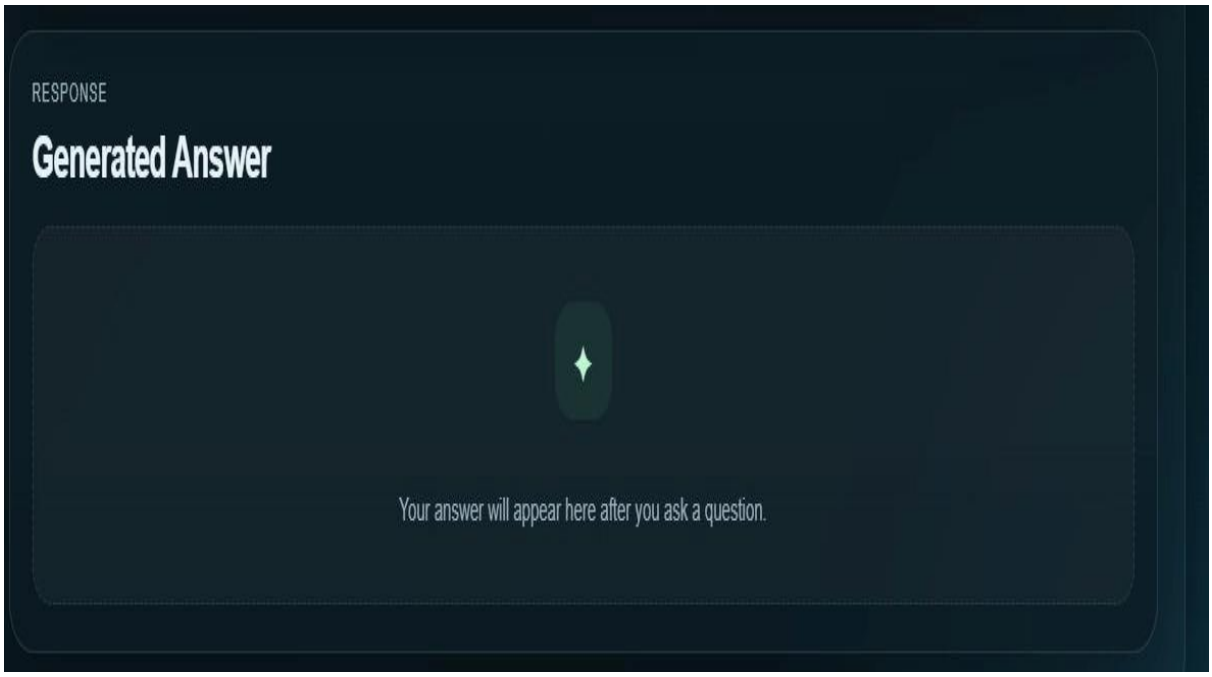
B.Uploading page



C. Questioning Page



D. Result Page



9. Conclusion

- The project successfully implements a **RAG-based intelligent document system**.
- It improves document search by using Semantic understanding Context-based retrieval
- The integration of vector databases and LLMs ensures High accuracy Efficient processing
- The system reduces dependency on manual document reading and analysis.
- It enhances productivity by providing **quick and reliable answers**.
- The project demonstrates the potential of **Generative AI in document intelligence systems**.

10. Future Scope & Enhancements

I. Support for Multiple File Formats

The system can be enhanced to support additional document formats such as Word (DOCX) and Excel (XLSX) along with PDFs. This will increase flexibility and allow users to process a wider variety of data sources commonly used in the energy sector.

II. Cloud Integration for Scalability

Integrating the system with cloud platforms like AWS or Azure will enable scalable storage and faster processing. It will also allow handling large volumes of documents and multiple users efficiently with improved system performance.

III. User Authentication and Security

Future improvements can include secure login systems and role-based access control. This ensures that only authorized users can access sensitive documents, making the system suitable for enterprise-level applications.

IV. Multilingual Support

Adding multilingual capabilities will allow users to upload documents and ask queries in different languages. This feature will improve accessibility and make the system usable across global organizations.

V. **Improved User Interface (UI/UX)**

Enhancing the user interface with better design, navigation, and interactive elements will improve usability. Features like dashboards, real-time feedback, and better visualization can significantly enhance user experience.

VI. **Fine-Tuning of LLM Models**

The accuracy of responses can be improved by fine-tuning the Large Language Model on domain-specific data related to the energy sector. This will help the system generate more precise and context-aware answers.

VII. **Voice-Based Query System**

Integration of speech recognition technology will allow users to interact with the system using voice commands. This will make the system more user-friendly and suitable for real-time application.

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